



CONTACT

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EDUCATION

Master of Science in AI & Robotics
Hof University of Applied Sciences
Mar 2025 – Mar 2027 (expected) · GPA 1.8
Generative AI, Machine Learning, Deep Learning, Data Mining, Advanced AI Applications, Predictive Maintenance & Condition Monitoring, Intelligent Robotics, Research Methodology & Academic Writing.

Bachelor of Science in Computer Science
Veer Narmad South Gujarat University
Aug 2021 – Mar 2024 · GPA 9.01
Data Structures & Algorithms, Object-Oriented Programming, Software Engineering, Database Management Systems, Web & Mobile Development, Mathematics (Calculus, Statistics, Linear Algebra).

CORE SKILLS

- LangGraph
- LangChain
- RAG
- Multi-Agent Systems
- Prompt Engineering
- LLM Evaluation
- Python
- FastAPI
- Pydantic
- Celery
- Redis
- REST APIs
- React
- Next.js
- TypeScript
- shadcn/ui

LANGUAGES

English — C1 (Business Fluent)
German — CEFR B2
Hindi, Gujarati — Native

CERTIFICATIONS

Oracle Agentic AI Certified Foundations Associate
Oracle · August 2026
Introduction to Model Context Protocol
Anthropic · March 2026

AWARDS

Dean's List
Hof University of Applied Sciences · 2025
Top academic performer
ML Hackathon 2.0 — 2nd Place
2023

Harsh Virani

AI Engineer · Agentic Systems, RAG & LLM Evaluation

PROFILE

Software engineer moving into applied AI: production LangGraph agent pipelines, retrieval-augmented generation, and LLM evaluation harnesses. One year shipping full-stack products, now focused on fault-isolated multi-agent systems with cost, latency, and groundedness instrumentation. M.Sc. AI & Robotics student in Germany, available as a Werkstudent.

EXPERIENCE

Software Engineer, AI & Full-Stack

Jul 2024 – Mar 2025

TechStaunch Solutions PVT LTD

- Built enterprise GenAI systems (RAG pipelines, chatbots, agentic workflows) with **LangChain**, **LlamaIndex**, and **FastAPI**, improving information-retrieval efficiency 35%.
- Delivered on-premise LLM adaptation and RAG pipelines that run with **no external API calls**, meeting enterprise privacy and data-residency constraints.
- Embedded AI into live client products on demand: a **video-snapshot generator** with automated transcript extraction, and an **automated lead-generation pipeline** for the company's own funnel, each replacing a manual workflow.
- Delivered full-stack features for a production e-commerce platform (**10K+ daily users**, **50K+ daily transactions**) and cut deployment time **87%** (8 h to 50 min) with an AWS CI/CD pipeline.

MASTER'S THESIS

Deep Research LLM-Agents for Office Work — An Empirical Investigation with Spreadsheet Software

M.Sc. thesis · Hof University · Supervisor: Prof. Dr. René Peinl · in progress

- Built an agent harness for spreadsheet work: a fixed set of actions (write formula, filter, sort, pivot, read range, validate) that each return structured feedback the model uses to correct itself over successive turns.
- Designed the execution environment as two containers on Kubernetes, one for the LLM and one running LibreOffice through its Python UNO bridge, network-isolated per task, so the user installs nothing locally.
- Kept large sheets out of the model's context (a few thousand rows already exceed 128k tokens) by running queries, filters, and aggregations inside the spreadsheet engine and passing back only the results.
- Designed two controlled experiments on SpreadsheetBench: first comparing about five open-weight self-hostable models on one fixed harness, then testing this harness against SWE-agent, Claude Code, and Cline on the strongest model, to separate the model's share of agent performance from the harness's (pass rate, cell- and task-level accuracy, effective-step ratio, cost, runtime).

SELECTED PROJECTS

Zero-Touch Customs Filing (Tdec)

Production 11-node LangGraph pipeline with browser-control and tool-calling agents

- Designed an **11-node LangGraph pipeline** (extract to track to validate to generate government XML) with a per-node guard decorator that degrades a bad PDF to a flagged job instead of a crashed worker, turning a **30-40 min** manual filing into a **~30-60 s** run.
- Added an **LLM-controlled headless-browser agent** (browser-use + Gemini + Playwright) that reads bot-protected FedEx/UPS/DHL portals automatically, plus a confidence-gated multimodal extraction stage that flags only uncertain fields, cutting the reviewer to a handful of checks instead of the whole 37-46-field form.
- Built a golden-set eval harness that runs the real pipeline with deterministic + LLM-judge scoring, driving field accuracy from **45.8% to ~88%** (F1 0.92) across prompt and pipeline iterations.

Tech: FastAPI, LangGraph, Celery, Redis, Gemini, browser-use, Next.js

TOOLS & PLATFORMS

PyTorchHugging Face

XGBoostscikit-learn

TensorFlowDockerCI/CD

AWSGCPVercel

PostgreSQLQdrantFAISS

LangfuseLangSmithSentry

pytest

AVAILABILITY

Available immediately

15-20 hours/week (semester), 30-40 hours/week (breaks)

Long-term through Master's program

CURRENT FOCUS

Multi-agent orchestration and fault isolation

LLM evaluation frameworks and reference benchmarks

Retrieval-augmented generation: hybrid search and reranking

Cost, latency and groundedness observability

Grounded-or-Silent Support Agent

6-node LangGraph graph, RAGAS-benchmarked at 0.958 faithfulness

- Architected a **6-node LangGraph agent** (classify to route to retrieve to generate to grade to retry) whose reflection loop rewrites its own query and re-retrieves on ungrounded answers, catching and reworking answers on **5 of 24** eval cases instead of shipping them.
- Built hybrid dense+sparse retrieval with RRF fusion and cross-encoder reranking over a ~20-candidate over-fetch, lifting RAGAS to **0.958 faithfulness / 0.889 context precision** on a 24-case set where dense-only retrieval missed exact-term queries.
- Fit the torch + sentence-transformers service onto a free-tier host: CPU-only PyTorch wheel and image-baked reranker models cut deploy size **~90%** (~2.5 GB to ~200 MB) and removed cold-start Hugging Face 429s so the container goes healthy immediately.

Tech: FastAPI, LangGraph, Qdrant, Neon Postgres, RAGAS, Render, Next.js

Agent Regression Radar

Trajectory-based evaluation and regression detection for LLM agents

- Designed a canonical **Agent IR** over OpenTelemetry GenAI spans (LangGraph instrumentor to ClickHouse) that reconstructs multi-turn sessions and each turn's tool/model steps, cutting multi-turn capture from N traces plus a cross-trace grouping pass to **one trace per conversation** and collapsing framework integration to a single adapter.
- Shipped **9 deterministic evaluators** (tool choice, tool arguments, JSON-schema, task-success, latency, cost, trajectory efficiency, tool-ordering) on one `evaluate(case, trajectory)` contract, so objective checks need **no LLM judge** and stay reproducible at zero API cost.
- Built uncertainty-aware aggregation (Wilson intervals for pass rates, bootstrap CIs for continuous metrics) that flags a version-to-version change as a regression only when it is both statistically credible and practically meaningful, removing raw-delta false alarms.

Tech: LangGraph, OpenTelemetry, ClickHouse, FastAPI, Pydantic v2, PostgreSQL, Next.js

RESEARCH

- Surveyed the evolving LLM ecosystem: decoder-only, encoder-only, and encoder-decoder architectures across NLU/NLG, covering prompt strategies, hallucination mitigation, and Code LLMs.
- Reviewed AI-augmented low-code / no-code platforms: LLM-based agents bridging engineers and citizen developers, task-technology fit, and workflow integration.